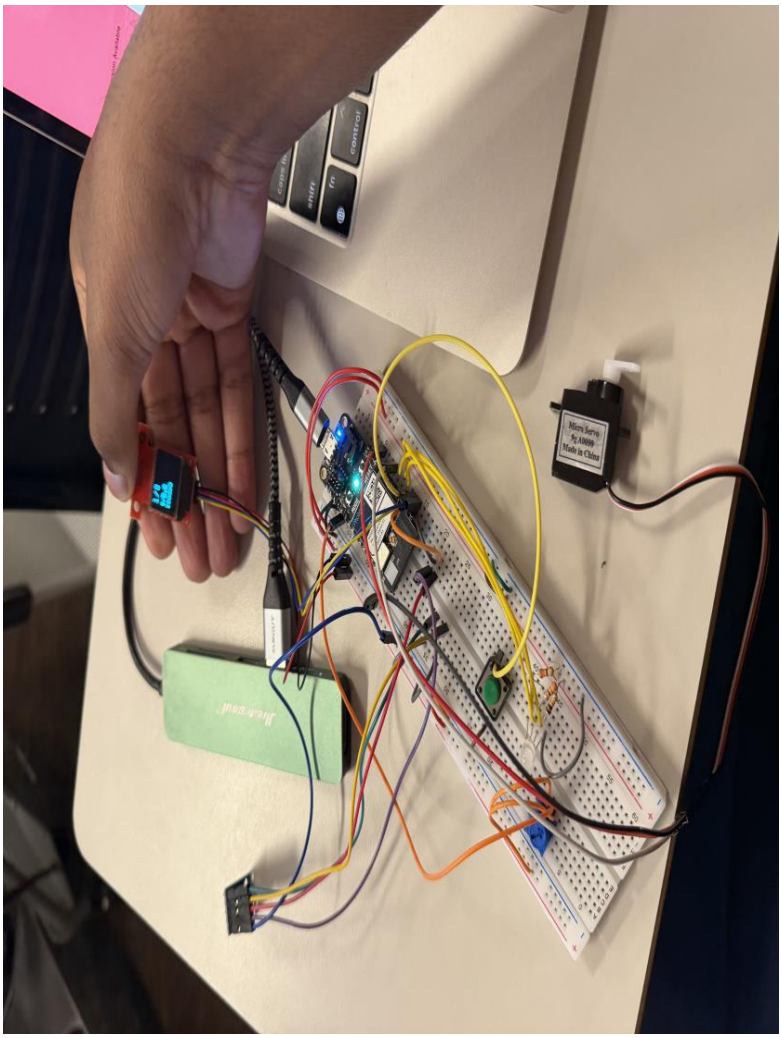
University of Southern
California TAC 348 - Making
Smart Devices Spring 2026

FINAL PROJECT: DEVELOPER DOCUMENTATION



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Author's Notes

Hi! Thanks for taking the time to look through this documentation. The Photon 2 Light Meter started as a way to bring digital convenience to film photography without giving up the slow, deliberate workflow that makes shooting film fun. Use this guide to understand how it's built and how to modify it for your own shooting style. Feel free to reach out with questions or suggestions.

OVERVIEW

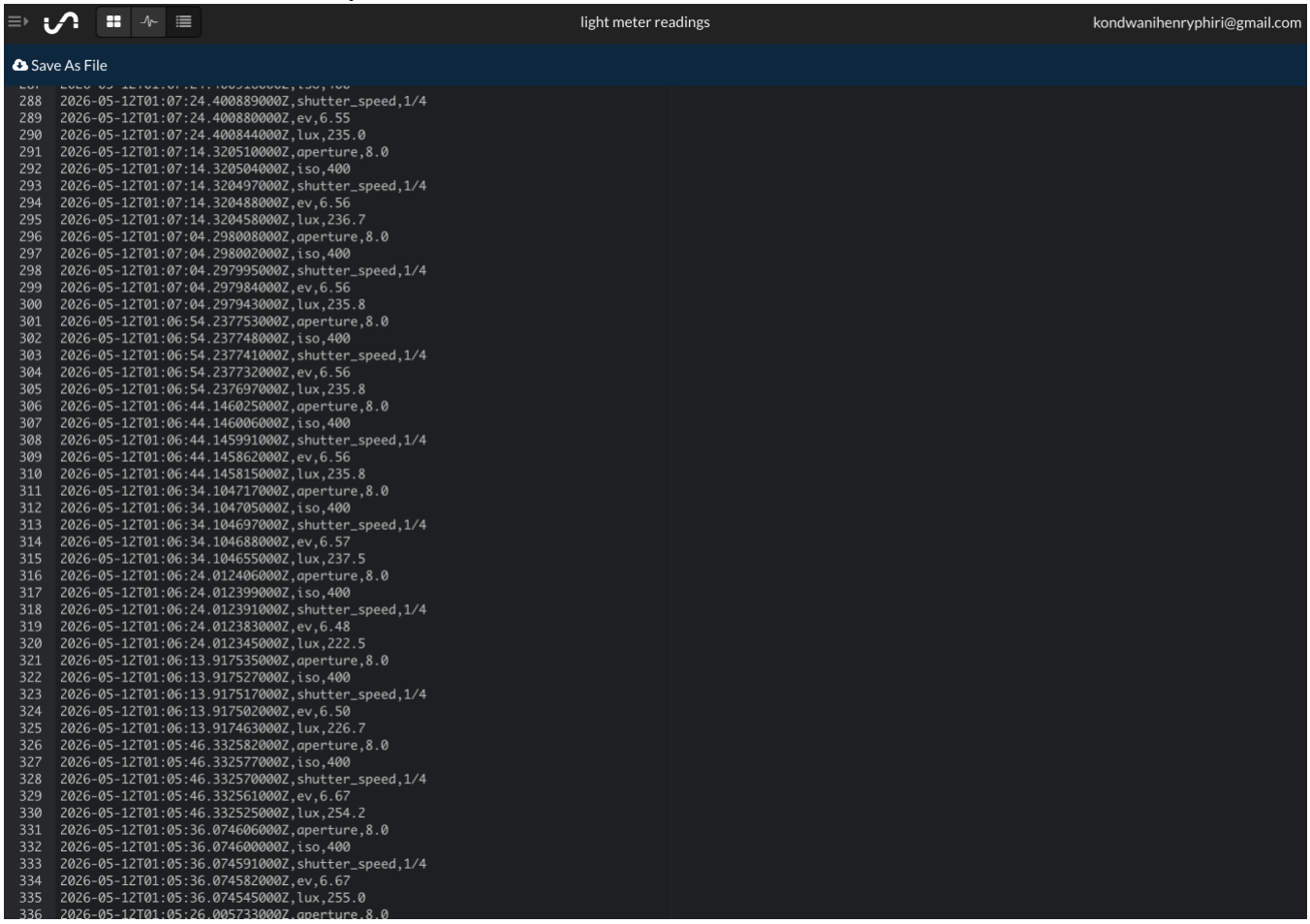
The Photon 2 Light Meter is a handheld light meter for film photography. It reads ambient light using a BH1750 digital sensor, calculates the Exposure Value (EV), and recommends a shutter speed based on the user's selected aperture and ISO. The reading appears on a small OLED screen, and a servo acts as a physical exposure needle to give the meter an analog feel.

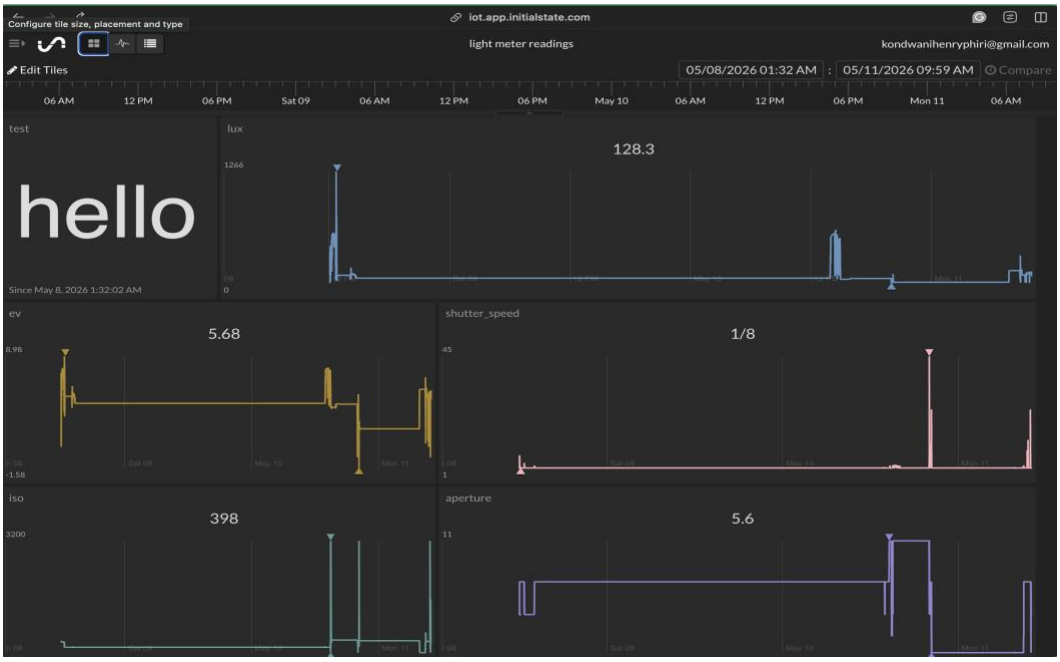
The user controls aperture with a potentiometer (dialing through standard f-stops from f/1.4 to f/16) and toggles between two display modes using a tactile button. An RGB LED indicates exposure status at a glance: green for correct, blue for underexposed, red for overexposed.

The **Blynk** mobile app is used to remotely set ISO from a phone, letting the user adjust the meter without touching the hardware.

Link to Blynk: https://blynk.cloud

All readings (lux, EV, recommended shutter speed, ISO, and aperture) are also streamed to an **InitialState** dashboard every 10 seconds, where the user can see live values and historical trends.




Configure tile size, placement and type
iot.app.initialstate.com
light meter readings
kondwanihenryphiri@gmail.com
Edit Tiles
05/08/2026 01:32 AM : 05/11/2026 09:59 AM
Compare
06 AM
12 PM
06 PM
Sat 09
06 AM
12 PM
06 PM
May 10
06 AM
12 PM
06 PM
Mon 11
06 AM
test
hello
Since May 8, 2026 1:32:02 AM
lux
128.3
1266
0
ev
5.68
8.98
-1.58
shutter_speed
1/8
45
1
iso
398
3200
aperture
5.6
11

WIRING DIAGRAM AND COMPONENTS

The wiring diagram below shows the layout used for the Photon 2 Light Meter. Wire colors and positions may vary slightly in physical implementation, but the pin assignments must stay the same.

Important: the servo must be powered from VUSB (5V), not the 3.3V rail. The 3.3V rail cannot deliver enough current for the servo to move. The OLED's reset pin lives on D8 to free up D6 for the servo signal.

Component	Purpose
BH1750 light sensor	Measures ambient light in lux
SparkFun Micro OLED	Displays EV, f-stop, shutter speed, and lux
Potentiometer	Lets the user dial in an f-stop value
Push button	Toggles between OLED display modes
RGB LED + 220□ resistor	Indicates exposure quality (over/under/good)
SG90 micro servo	Physical exposure needle (analog feel)
Particle Photon 2	Microcontroller with built-in Wi-Fi

Pin assignments on the Photon 2:

Photon 2 Pin	Connected To
3.3V	BH1750 VCC, OLED VCC, potentiometer right leg
VUSB (5V)	Servo red wire
GND	All component grounds
D0 (SDA)	BH1750 SDA + OLED SDA (shared I2C bus)
D1 (SCL)	BH1750 SCL + OLED SCL (shared I2C bus)
A0	Potentiometer wiper
D2	Button (other leg to GND)
D3 / D4 / D5	RGB LED blue / green / red legs (via 220□)
D6	Servo signal wire
D8	OLED reset pin

DEVICE SETUP INSTRUCTIONS

1. Wire all components according to the wiring diagram. Confirm the servo's red wire is on VUSB, not 3.3V.

2. Open Visual Studio Code and use the Command Palette: **Particle: Configure Project for Device**. Select **Photon 2** and **Device OS 6.2.1**.

3. Make sure project.properties lists these dependencies: SparkFunMicroOLED=1.3.0, BH1750=1.0.0, blynk=0.6.1. Workbench will fetch them on first build.

4. Set up an InitialState bucket and copy the bucket key (e.g. 54P9ETXP8KXW) and your streaming access key.

5. In Particle Console, create a webhook called **lightmeter_data** pointing to https://groker.init.st/api/events. Add the InitialState headers and a JSON body that maps the event fields to InitialState keys.

6. Set up a Blynk template called **Light Meter** with a single integer datastream on **V0** for ISO (min 100, max 3200, default 400). Create a device from the template and copy the three credentials (template ID, template name, auth token).

7. Paste your Blynk credentials at the top of src/check_oled_check_sensor.cpp. They must come **before** the #include "blynk.h" line.

8. Add a Slider widget in the Blynk mobile app, set its datastream to ISO (V0), min 100 / max 3200.

9. Flash the project locally using the lightning bolt in the VS Code status bar.

10. Once the Photon is breathing cyan, the OLED will display the current EV, f-stop, shutter speed, and lux reading. Press the button to toggle to the simplified view with a large shutter speed.

After setup, both the Blynk slider and the InitialState dashboard should be live. Sliding the ISO control in the app updates the device in real time, and all five readings appear on the dashboard within 10 seconds.

SUMMARY OF CODE

The firmware is contained in a single file: src/check_oled_check_sensor.cpp. It is organized into the following sections (labeled with comments in the code).

Section 1: Blynk credentials and includes

The Blynk template ID, template name, and auth token are defined at the top of the file using #define. These must come before the #include "blynk.h" line or the build will fail. Library includes follow.

Section 2: Pin definitions and constants

All pin assignments live here as #define macros so they're easy to find and change. Also defines the cloud push interval (10 seconds), button debounce delay, and servo timing constants.

Section 3: Global state

Tracks aperture, ISO, display mode, button debounce state, servo angle and direction, and timestamps for the cloud push timer and servo step timer.

Section 4: Helper functions

Small utility functions: setLED(r, g, b) drives the RGB LED, mapAperture(raw) buckets the potentiometer's 0-4095 reading into 8 standard f-stops.

Section 5: Sensor and exposure math

readLux() reads the BH1750 over I2C. luxToEV(lux) converts lux to Exposure Value using the formula EV = log■(lux / 2.5). getShutterString(lux) computes the recommended shutter time from EV, aperture, and ISO, then snaps the result to the nearest standard shutter speed (1/8000 through 2s).

Section 6: Blynk handler

BLYNK_WRITE(V0) fires automatically whenever the ISO slider changes on the phone. It updates the global ISO variable, which the next loop iteration uses to recompute the shutter speed.

Section 7: Servo sweep

updateServoSweep() non-blockingly steps the servo angle by 5 degrees every 20ms, bouncing between 0° and 180°. The continuous motion acts as a heartbeat indicator: as long as the needle is moving, the light meter is alive and reading data.

Section 8: Cloud push

pushToInitialState(...) builds a JSON payload with all 5 readings and calls Particle.publish("lightmeter_data", ...). The webhook in Particle Console catches this event and forwards it to InitialState over HTTPS.

Section 9: Setup

Initializes serial, pin modes, the servo, the BH1750, and the OLED. Waits up to 10 seconds for the Particle cloud connection, then starts Blynk.

Section 10: Main loop

Runs the Blynk client, reads the potentiometer and button, reads lux from the BH1750, computes EV and shutter speed, updates the LED based on exposure range, advances the servo sweep, redraws the OLED, and pushes data to InitialState on a 10-second timer. A 50ms delay at the end of the loop keeps the servo sweep smooth without saturating the I2C bus.

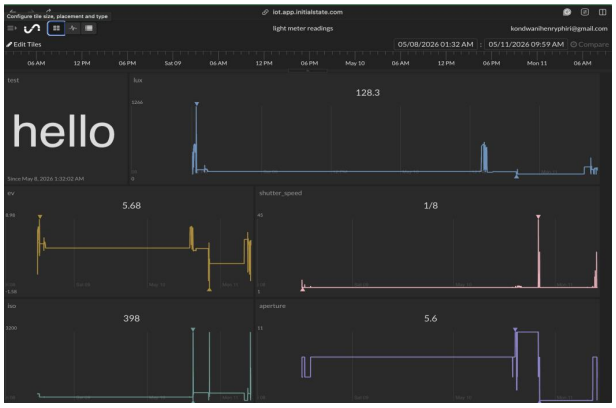
CLOUD VARIABLES AND DASHBOARD

This project uses two cloud services: **InitialState** for the dashboard and **Blynk** for remote control. The Particle webhook integration bridges the device to InitialState over HTTPS.

The dashboard logic is in the main loop, where a millis() timer fires Particle.publish() every 10 seconds. The published event is named **lightmeter_data** and contains a JSON payload with all five readings.

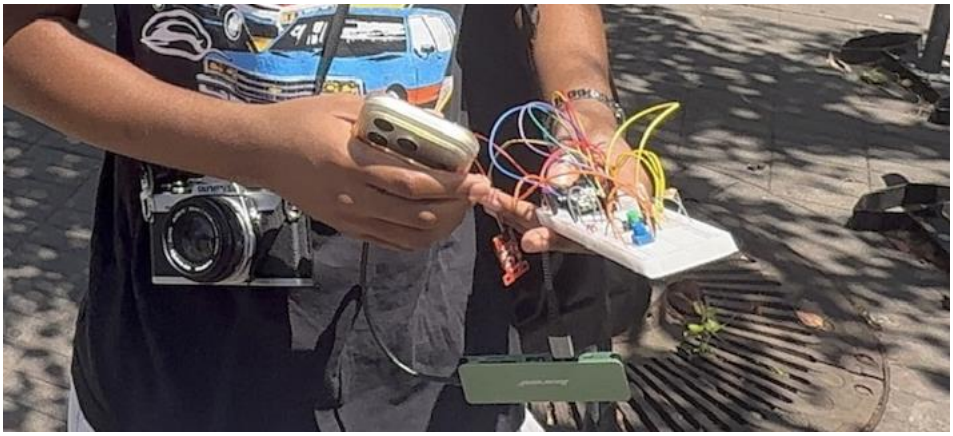
Stream	Description
lux	Raw ambient light reading from the BH1750 (0-65,535 lux)
ev	Calculated Exposure Value (typically 1-14 for usable scenes)
shutter_speed	Recommended shutter speed, snapped to standard values
iso	Current ISO setting (controlled remotely via Blynk)
aperture	Current f-stop selected via potentiometer (1.4-16)

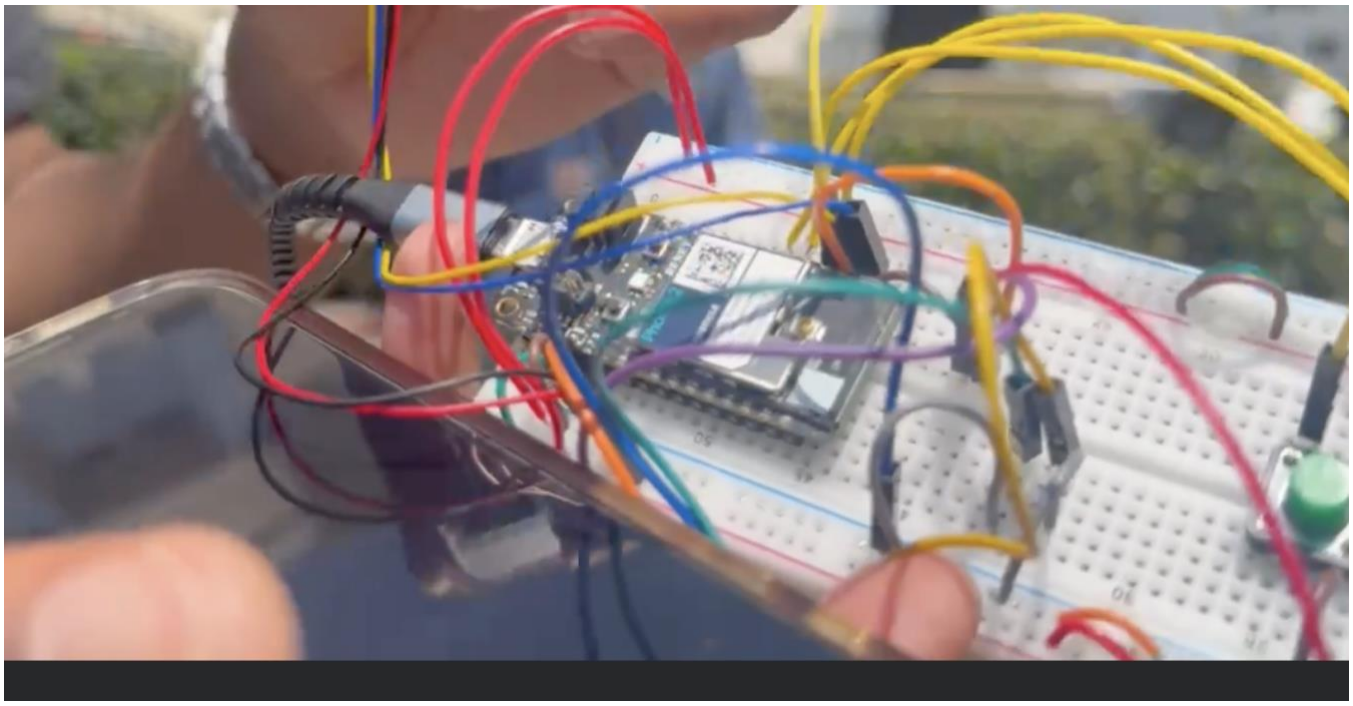
Blynk handles remote control via a single Virtual Pin (**V0**) that's mapped to a slider in the mobile app. The slider's range is 100-3200 (covering common film ISO values), with a default of 400.



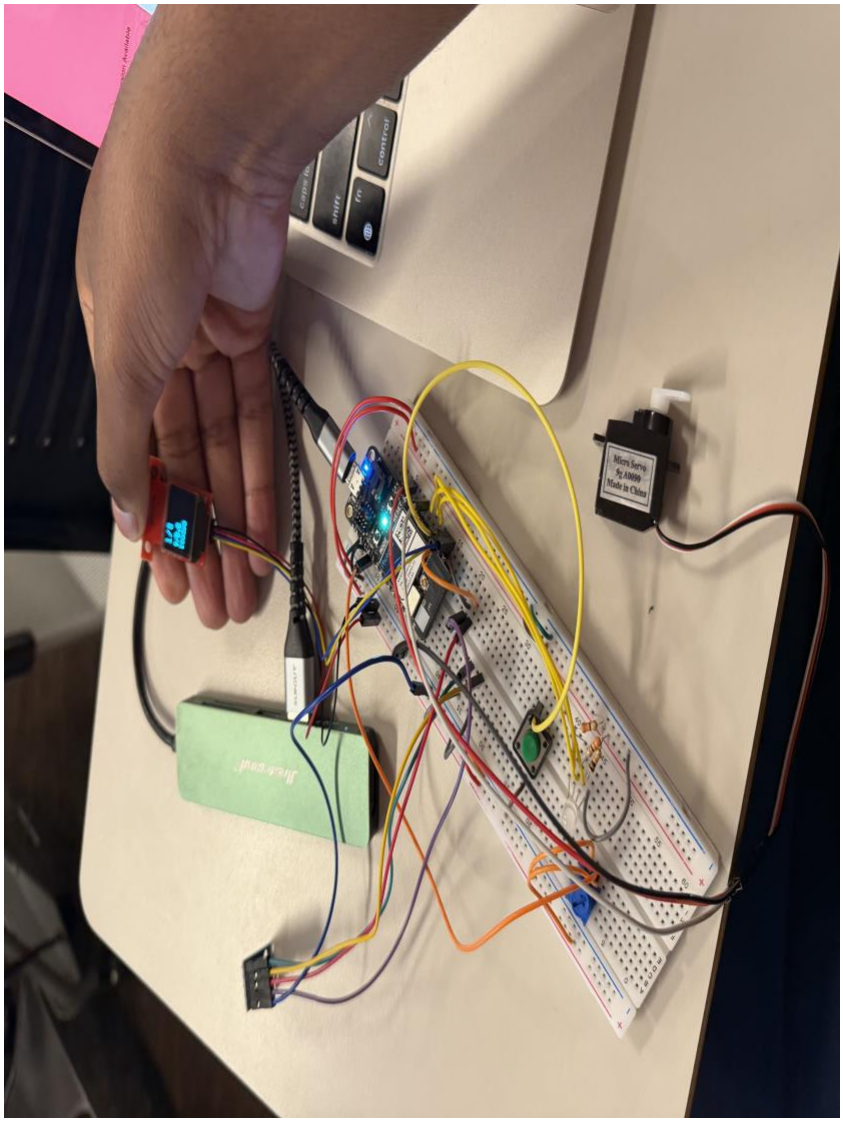


PHOTOS






EV 4
f/5.6
1" ISO400
78 1x



Thanks for reading. Please reach out with any questions or suggestions.